

Pairing Proofs and Polynomial Ring Relations

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Abstract

In this work, we consider the setting where elliptic curve pairings need to be performed within another proving system. This appears in proof recursion where a proof verifies other proofs, or when pairing-based protocols like BLS (Boneh-Lynn-Shacham) signatures or KZG commitments are used within a proof. We show a construction that simultaneously reduces arithmetic circuit depth and communication complexity.

We present *a two-round public-coin interactive proof system* that consolidates the Miller loop iterations and final exponentiation into unified polynomial relations, verified as polynomial identities over $\mathbb{F}_q[x]$. This construction significantly reduces computational overhead across R1CS, Plonkish and AIR-based systems. We prove knowledge soundness $\delta_s \leq 2^{-245}$ for a three-pair optimal ate multi-pairing on BN254 curve.

As our *second contribution* we present a generalised construction of our core technique, parameterised by an arbitrary modulus polynomial, designed for easy application across various protocols and schemes beyond pairings that utilize polynomial ring multiplications.

We demonstrate the practical efficacy of our construction across R1CS, Plonkish, and AIR arithmetisations on BN254 and BLS12-381, with implementations in gnark—the open-source zk-SNARK ecosystem—and in Garaga, the standard library for pairing-based ZK proof verification on Cairo, a leading AIR-based CPU architecture.

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1 Introduction

Bilinear pairings on elliptic curves are a fundamental building block of modern cryptography, underlying SNARK constructions such as Groth16 [Gro16] and PLONK [GWC19], the KZG polynomial commitment scheme [KZG10], and BLS signatures [BGW⁺22]. While native pairing computation is well-studied and highly optimised, a different challenge arises when a pairing must be verified *inside another proving system*—that is, when the pairing computation itself becomes a statement to be proved in an arithmetised circuit.

This setting arises naturally in **proof recursion**, where one SNARK verifies another, and more broadly whenever a verifier is itself constrained: in resource-metered execution environments, in on-chain verification, and in any arithmetised computation system (R1CS [GGPR13], Plonkish [GWC19], AIR [GPR21]) where each $\mathbb{F}_{q^{12}}$ operation translates into many constraints. In these contexts, the cost of expressing a pairing as a circuit dominates verification, and reducing it has been the subject of a substantial line of work.

1.1 Our Contribution

We present a public-coin interactive proof system for verifying pairing computations that consolidates verification into unified polynomial relationships under the Schwartz-Zippel lemma. By treating complete Miller loop iterations as single polynomial relationships rather than sequences of individual extension field operations [Fel23], our approach achieves:

- **39% reduction** in Plonkish constraints ($1,890,771 \rightarrow 1,158,194$ for BN254 Groth16 simulation)
- **60% reduction** in communication complexity ($739,198 \rightarrow 295,946$)
- **~42% reduction** in compile-time memory and up to **52%** in solver memory

Our key insight is that instead of verifying individual field operations separately, we can verify entire Miller loop iterations as single polynomial identities. This consolidation dramatically reduces both the number of modular operations required and the communication overhead from intermediate commitments.

As a reusable artefact of independent interest, we also contribute a polynomial ring toolkit for gnark’s emulated arithmetic package. The toolkit instantiates the two-round interactive proof system generically: it is parameterised by an arbitrary modulus polynomial $P \in \mathbb{F}_q[x]$ and an emulated base field, so the same construction applies to pairing verification on any curve and, more broadly, to any computation that reduces to multiplications in $\mathbb{F}_q[x]/\langle P \rangle$. Multiplications submitted to the toolkit are verified lazily:

the prover’s quotient and remainder are computed outside the circuit, and the resulting polynomial-identity check is deferred to circuit finalisation, where all checks under a common modulus are batched into a single Schwartz–Zippel test against two Fiat–Shamir challenges.

1.2 Related Work

The landscape of efficient pairing verification in constraint systems has evolved through three key contributions that our work directly builds upon.

El Housni’s **Pairings in Rank-1 Constraint Systems** [Hou23] established the theoretical framework for optimizing pairings in arithmetized environments, introducing efficient Miller loop formulas in affine coordinates and sparse line function representations. This work inverts the usual cost model: inversions become cheap (2C) while multiplications remain the dominant cost.

Feltroidprime [Fel23] introduced polynomial identity testing for $\mathbb{F}_{q^{12}}$ arithmetic via the Schwartz–Zippel lemma. Each multiplication $A \cdot B$ is expressed as $A(x) \cdot B(x) = Q(x) \cdot P_{12}(x) + R(x)$ and verified at a single random point, with the quotients Q batched via random linear combination. Our work applies the same polynomial framework but at a coarser granularity: entire Miller loop steps rather than individual multiplications, eliminating intermediate remainder commitments entirely.

Novakovic and Eagen’s **On Proving Pairings** [NE24] showed that final exponentiation can be replaced by a residue witness check, reducing final exponentiation cost by over 85% for BN254. We integrate their residue witness c directly into our polynomial step equations (Section 3.2.2), unifying the Miller loop and final exponentiation into the same polynomial verification framework.

These works provide the baseline for our construction. Our work introduces optimizations that further reduce constraint counts beyond the cumulative gains of these existing techniques.

2 Background

2.1 Pairing Computation

An elliptic curve pairing is a non-degenerate bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, where $\mathbb{G}_1$ and $\mathbb{G}_2$ are groups of points on elliptic curves over finite fields, and $\mathbb{G}_T$ is a multiplicative group in an extension field. For practical cryptographic applications, we require that this map be efficiently computable. We will focus on the optimal ate pairing [BGM⁺10] over Barreto–Naehrig (BN) curves, which is widely used due to its efficiency. We will specifically use the Ethereum’s BN254 curve, defined over a 254-bit prime field $\mathbb{F}_q$, with groups $\mathbb{G}_1, \mathbb{G}_2$ of order r and target group $\mathbb{G}_T$ contained in the 12th extension field $\mathbb{F}_{q^{12}}$.

The computation of an optimal ate pairing—the focus of our implementation—consists of two primary phases:

1. **Miller loop:** The core iterative computation that evaluates line functions at pairs of points from $\mathbb{G}_1$ and $\mathbb{G}_2$, accumulating intermediate results to produce a value in $\mathbb{F}_{q^{12}}$.

Algorithm 1 Optimal Ate pairing over Barreto-Naehrig curves [BGM⁺10]

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$ **Output:** $a_{\text{opt}}(Q, P)$

```
1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $T \leftarrow Q, f \leftarrow 1$ 
3: for  $i = L - 2$  to  $0$  do
4:    $f \leftarrow f^2 \cdot l_{T,T}(P)$ 
5:    $T \leftarrow [2]T$ 
6:   if  $s_i = 1$  then
7:      $f \leftarrow f \cdot l_{T,Q}(P)$ 
8:      $T \leftarrow T + Q$ 
9:   end if
10:  if  $s_i = -1$  then
11:     $f \leftarrow f \cdot l_{T,-Q}(P)$ 
12:     $T \leftarrow T - Q$ 
13:  end if
14: end for
15:  $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 
16:  $f \leftarrow f \cdot l_{T,Q_1}(P), T \leftarrow T + Q_1$ 
17:  $f \leftarrow f \cdot l_{T,-Q_2}(P), T \leftarrow T - Q_2$ 
18:  $f \leftarrow f^{(p^{12}-1)/r}$ 
19: return  $f$ 
```

2. **Final Exponentiation:** This step raises the Miller loop output to the power $(q^{12} - 1)/r$, where r is the order of the groups, ensuring the result lies in the correct subgroup $\mathbb{G}_T$.

While both phases present optimization opportunities, the constraint-based verification of these computations introduces unique challenges and possibilities that differ significantly from native implementations.

2.2 Schwartz-Zippel lemma

The Schwartz-Zippel lemma (or more precisely, the Demillo-Lipton-Schwartz-Zippel lemma [DL78, Sch79, Zip79, Sch80]) provides a probabilistic method for testing polynomial identities. The lemma states that for a non-zero polynomial $P(x_1, x_2, \dots, x_n)$ of total degree d over a finite field $\mathbb{F}$, when evaluated at uniformly random field elements, the probability that P evaluates to zero is at most $d/|\mathbb{F}|$.

This probabilistic bound becomes negligible for cryptographically sized fields where $d \ll |\mathbb{F}|$, making it practical to verify polynomial identities by evaluation instead of coefficient-wise operations.

2.3 Computation Models

We implement and benchmark our toolkit in gnark [Cona], the industry-standard Go library for arithmetized circuit development, targeting both SCS (Sparse Constraint System, the native Plonkish backend) and R1CS constraint systems. We map each field multiplication to a constraint (denoted by **C**) to maintain compatibility with the R1CS

model used in established literature. In R1CS [Gro16], prover complexity is dominated by multiplication gates while additions are free. To synchronize our theoretical analysis with the benchmarks provided by El Housni [Hou23], we define our cost metric based on field multiplications.

2.3.1 R1CS Cost Model

We measure computational cost in terms of *constraints* (denoted by $\mathbf{C}$), representing the fundamental unit of work in Rank-1 constraint systems. One constraint corresponds to one multiplication gate.

Basic Field Operations in $\mathbb{F}_q$:

- **Addition/Subtraction:** Free
- **Multiplication:** One constraint (1C)
- **Division/Inversion:** One multiplication + one equality check

Commitment Overhead: Beyond constraint costs, we account for *calldata* or *commitment overhead*—the number of field elements the prover must commit to (via Fiat-Shamir hashing in non-interactive proofs). For blockchain deployments, this overhead directly impacts transaction costs and is often the dominant factor.

2.3.2 Cost Model Comparison

Operations that are traditionally expensive (inversions, divisions) become cheap in constraint systems when computed via witnesses. This inverts traditional optimization priorities: we can afford inversions to save multiplications, enabling affine coordinate systems and other techniques impractical in native implementations.

Recognizing and exploiting these unique computational properties is essential for unlocking hidden performance potential. Youssef El Housni’s pioneering work **Pairings in Rank-1 Constraint Systems** [Hou23] represents, to our knowledge, the first systematic exploration of these optimization opportunities, providing a breakthrough in the practicality of recursive proving by thoughtfully leveraging the distinctive cost model inherent to constraint-based systems.

3 Pairing Operations as Polynomials

Algorithm ?? gives the pairing computation combining El Housni’s R1CS optimizations [Hou23] with Novakovic-Eagen’s residue witness technique [NE24]. The 27th root of unity W scales the Miller loop output f to be a cubic residue (see [NE24] §4.3); if f is not already a cubic residue, multiplying by W^w , $w \in \{1, 2\}$ corrects this.

Our contribution is to express each step of this algorithm as a single multi-operand polynomial product verified via the Schwartz-Zippel lemma, rather than as a sequence of individual binary multiplications as in [Fel23].

3.1 Comparative Analysis

We compare the two schemes for a 3-pair Miller loop zero-bit step (the most common case in BN254’s NAF [DHM04] representation of $s = 6t + 2$). Here $F \in \mathbb{F}_{q^{12}}$ is the

Algorithm 2 Optimal Ate Multi-Pairing with Residue witness check

Input: Vectors $\vec{P} = (P_1, \dots, P_N) \in \mathbb{G}_1^N$, $\vec{Q} = (Q_1, \dots, Q_N) \in \mathbb{G}_2^N$, $c \in \mathbb{F}_{q^{12}}$, $w \in \{0, 1, 2\}$
 Internal $W, W^2 \in \mathbb{F}_{q^{12}}$ $\triangleright W$ is 27th root of unity

Output: $\prod_{k=1}^N a_{opt}(Q_k, P_k)$

```

1: Write  $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$  where  $s_i \in \{-1, 0, 1\}$  and  $s_L = 1$ 
2:  $c^{-1} \leftarrow 1/c$   $\triangleright$  Inverse Residue witness
3:  $f \leftarrow c^{-1}$   $\triangleright$  Initialize with inverse residue witness
4: for  $k = 1$  to  $N$  do  $T_k \leftarrow Q_k$ 

5: for  $i = L - 2$  to  $0$  do
6:    $f \leftarrow f^2$ 
7:    $f \leftarrow f \cdot \prod_{k=1}^N l_{T_k, T_k}(P_k)$ 
8:   for  $k = 1$  to  $N$  do  $T_k \leftarrow [2]T_k$ 
9:   if  $s_i = 1$  then
10:     $f \leftarrow f \cdot c^{-1}$   $\triangleright$  Residue witness
11:     $f \leftarrow f \cdot \prod_{k=1}^N l_{T_k, Q_k}(P_k)$ 
12:    for  $k = 1$  to  $N$  do  $T_k \leftarrow T_k + Q_k$ 
13:  else if  $s_i = -1$  then
14:     $f \leftarrow f \cdot c$   $\triangleright$  Residue witness
15:     $f \leftarrow f \cdot \prod_{k=1}^N l_{T_k, -Q_k}(P_k)$ 
16:    for  $k = 1$  to  $N$  do  $T_k \leftarrow T_k - Q_k$ 
17:  end if
18: end for

19: for  $k = 1$  to  $N$  do  $Q_{1,k} \leftarrow \pi_p(Q_k)$ ,  $Q_{2,k} \leftarrow \pi_p^2(Q_k)$ 
20:  $f \leftarrow f \cdot \prod_{k=1}^N l_{T_k, Q_{1,k}}(P_k)$ 
21: for  $k = 1$  to  $N$  do  $T_k \leftarrow T_k + Q_{1,k}$ 
22:  $f \leftarrow f \cdot \prod_{k=1}^N l_{T_k, -Q_{2,k}}(P_k)$ 

23: if  $w = 1$  then
24:    $f \leftarrow f \cdot W$ 
25: else if  $w = 2$  then
26:    $f \leftarrow f \cdot W^2$ 
27: end if
28:  $f \leftarrow f \cdot c^{-q} \cdot c^{q^2} \cdot c^{-q^3}$   $\triangleright$  Uses Frobenius maps, Negative exponents use  $c^{-1}$ 

29: return  $f$ 

```

accumulator, $L_i \in \text{Sparse}_{01379}$ are the three doubling line functions, $R \in \mathbb{F}_{q^{12}}$ is the updated accumulator, and Q is the quotient polynomial. Sparse_{01379} denotes the subspace of $\mathbb{F}_{q^{12}}$ elements whose only nonzero coefficients are at degrees 0, 1, 3, 7, 9—the form that BN254 line functions naturally take [Hou23], reducing their evaluation cost from 12C to 4C.

Adapting [Fel23] to lines 6–7 of Algorithm ?? chains four binary multiplications, each producing an intermediate remainder that must be committed:

$$\begin{aligned} F(x) \cdot F(x) &\stackrel{?}{=} Q_1(x) \cdot P_{12}(x) + T_1(x) \\ T_1(x) \cdot L_1(x) &\stackrel{?}{=} Q_2(x) \cdot P_{12}(x) + T_2(x) \\ T_2(x) \cdot L_2(x) &\stackrel{?}{=} Q_3(x) \cdot P_{12}(x) + T_3(x) \\ T_3(x) \cdot L_3(x) &\stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x) \end{aligned}$$

Each of the five polynomial evaluations ($F, T_1\text{--}T_3, R$) costs 12C and each line costs 4C, with all four remainders requiring commitment: **76C and 48 $\mathbb{F}_q$ commitments**.

Our scheme collapses this into a single identity:

$$F(x)^2 \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) \stackrel{?}{=} Q(x) \cdot P_{12}(x) + R(x)$$

Only R requires commitment, and only F and R need full 12C evaluation; the three sparse lines cost 4C each: **41C and 12 $\mathbb{F}_q$ commitments** — a **46% constraint reduction** and **75% fewer commitments**. The comparison across all three schemes is summarised in Table 1.

Table 1: 3-pair Miller loop zero-bit step comparison

Scheme	Constraints	$\mathbb{F}_q$ Commitments	vs baseline
[Hou23]	132C	—	—
[Fel23]	76C	48	−42%
Ours	41C	12	−69%
[Hou23]: $36\text{C} + 3 \times 30\text{C} = 132\text{C}$			
[Fel23]: $5 \times 12\text{C} + 3 \times 4\text{C} + 4\text{C} = 76\text{C}$			
Ours: $2 \times 12\text{C} + 3 \times 4\text{C} + 5\text{C} = 41\text{C}$ (excl. Fiat-Shamir overhead)			

3.2 Operation Polynomials

We now state the verification equation for each step of Algorithm ?? in a 3-pairing configuration (as used in Groth16 verification). All equations share the form $\text{LHS} = R(x) + Q(x) \cdot P_{12}(x)$; the quotient Q is amortized across all steps via batching (Section 4.1).

3.2.1 Zero-bit step

Lines 6–7: one squaring and three doubling lines.

$$F^2(x) \cdot L_1(x) \cdot L_2(x) \cdot L_3(x) = R(x) + Q(x) \cdot P_{12}(x) \tag{1}$$

$F, R \in \mathbb{F}_{q^{12}}$ (12 coefficients each); $L_1, L_2, L_3 \in \text{Sparse}_{01379}$ (4 coefficients each). **Cost: 41C, 12 commitments.**

3.2.2 Non-zero bit step

Lines 10–15: squaring, residue witness, and both doubling and addition lines for each pair.

$$F^2(x) \cdot W^{\pm 1}(x) \cdot L_{d1}(x) \cdot L_{d2}(x) \cdot L_{d3}(x) \cdot L_{a1}(x) \cdot L_{a2}(x) \cdot L_{a3}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (2)$$

W^{-1} is used for negative bits; the equation structure is identical. $W \in \mathbb{F}_{q^{12}}$ encapsulates the equivalence class relationship from [NE24]. **Cost: 69C, 12 commitments.**

3.2.3 Frobenius correction step

Lines 20–22: no squaring; two sparse correction lines per pair for the π_p and $-\pi_p^2$ mappings.

$$F(x) \cdot L_{1\pi}(x) \cdot L_{2\pi}(x) \cdot L_{3\pi}(x) \cdot L_{1\pi^2}(x) \cdot L_{2\pi^2}(x) \cdot L_{3\pi^2}(x) = R(x) + Q(x) \cdot P_{12}(x) \quad (3)$$

Cost: 55C, 12 commitments.

3.2.4 Final exponentiation step

Lines 24–28: cubic scaling and the Frobenius component $(q - q^2 + q^3)$ of the final exponentiation, following [NE24]. $R = 1$ for a valid pairing.

$$F(x) \cdot c^{-q}(x) \cdot c^{q^2}(x) \cdot c^{-q^3}(x) \cdot W^w(x) = 1 + Q(x) \cdot P_{12}(x) \quad (4)$$

$w \in \{0, 1, 2\}$ selects the cubic root of unity scaling; W^w is sparse with 2 nonzero coefficients. The Frobenius images of c are provided as prover inputs. **Cost: 56C, 12 commitments.**

3.3 Security

Each verification equation takes the form $P = \prod_i A_i(x) - Q(x) \cdot P_{12}(x) - R(x)$ where $\deg(P) < 2^8$.¹ By the Schwartz-Zippel lemma (Section 2.2):

Soundness: $\delta_s \leq d/|\mathbb{F}_q| < 2^8/2^{254} = 2^{-246}$ for BN254.

Completeness: An honest prover always produces $P(x) = 0$ identically, so the verifier always accepts. Euclidean division guarantees unique Q, R with $\deg(R) < \deg(P_{12}) = 12$.

4 Full Pairing Proof with Quotient Batching

We now combine the polynomial equations from Section 3.2 into a complete two-round public-coin interactive proof, further reducing verification cost by batching all quotients into a single accumulated polynomial.

¹For k -pair pairings, $\deg(P) \approx 88 + 18(k - 3)$, well below 2^8 for all practical k . This bound can be extended to 2^{10} with negligible soundness impact.

4.1 Batching Polynomial Identity Testing

All n polynomial equations from Section 3.2 are batched into a single verification. The prover commits to remainders $\{R_i\}_{i=0}^{n-1}$ and receives challenge z . Using z , the prover forms the accumulated quotient $Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$. The verifier then samples a second challenge point c and checks:

$$\sum_{i=0}^{n-1} z^i \cdot [A_i(c) \cdot B_i(c) \cdots X_i(c) - R_i(c)] = Q_{\text{acc}}(c) \cdot P_{12}(c) \quad (5)$$

This reduces n polynomial verifications (degrees 38–76, see Tables ??–??) to a single evaluation of a degree-76 polynomial.

4.2 Prover’s Algorithms

The prover computes quotients Q_i and remainders R_i for each equation in Section 3.2. For each step, it calls **POLYMULTDIV** (Algorithm 3), which multiplies the input polynomials and performs standard Euclidean division by the irreducible P_{12} . After receiving challenge z , the prover forms $Q_{\text{acc}} = \sum_i z^i Q_i$ and sends $\mathcal{R} = \{R_i\}$ and Q_{acc} to the verifier. The full prover procedure is given in Algorithm 4.

Algorithm 3 PolyMulDiv: Multiply polynomials and divide by P_{12}

Input: $P_1, \dots, P_n \in \mathbb{F}_q[x]$; Divisor $P_{12} \in \mathbb{F}_q[x]$

Output: Q, R s.t. $\prod P_i = Q \cdot P_{12} + R$, $\deg(R) < \deg(P_{12})$

- 1: $R \leftarrow \prod_{i=1}^n P_i$ ▷ Polynomial product
 - 2: Perform Euclidean division of R by P_{12} to get (Q, R)
 - 3: **return** (Q, R)
-

4.3 Verifier’s Algorithms

The verifier receives $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$, residue witness c [NE24], the W (27th root of unity), the remainder array $\mathcal{R}$, and Q_{acc} from the prover. Using challenge z from the first round and a freshly sampled x , it mirrors the prover’s pairing loop while evaluating each polynomial at x via Algorithm 5, accumulating the RLC e_{acc} . The final check is Equation 5. The full verifier is given in Algorithm 6.

Algorithm 4 Pairing Prover: Preparing polynomials

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}$ Internal: $W, W^2 \in \mathbb{F}_{q^{12}}$ $\triangleright W$ is 27th root of unity**Output:** Arrays $\mathcal{Q}, \mathcal{R}$ of quotients and remaindersWrite $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$ 1: $c^{-1} \leftarrow 1/c$ $\triangleright$ Inverse Residue witness2: $f \leftarrow c^{-1}$ $\triangleright$ Initialize with inverse residue witness3: $T \leftarrow Q$ $\triangleright T_i$ for each pair for multi-pairingEvery operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.

Miller loop bits

4: **for** $i = L - 2$ to 0 **do**5: $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$ 6: **if** $s_i = 0$ **then**7: $Q, R \leftarrow \text{PolyMulDiv}(f, f, L_{\text{dbl}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i 8: **else if** $s_i = 1$ **then**9: $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$ 10: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i 11: **else if** $s_i = -1$ **then**12: $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$ 13: $Q, R \leftarrow \text{PolyMulDiv}(f, f, c, L_{\text{dbl}}, L_{\text{add}})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i 14: **end if**15: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$ 16: **end for**

Miller loop Frobenius correction

17: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$ 18: $L_\pi \leftarrow l_{T,Q_1}(P)$ 19: $T \leftarrow T + Q_1$ 20: $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$ 21: $Q, R \leftarrow \text{PolyMulDiv}(f, L_\pi, L_{-\pi^2})$ $\triangleright$ Algorithm 3, add L_{*i} for each P_i, Q_i 22: $f \leftarrow R, \mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$

Final exponentiation: Cubic scaling + Frobenius mappings

23: **if** $w = 0$ **then**24: $Q, R \leftarrow \text{PolyMulDiv}(f, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 325: **else if** $w = 1$ **then**26: $Q, R \leftarrow \text{PolyMulDiv}(f, W, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 327: **else if** $w = 2$ **then**28: $Q, R \leftarrow \text{PolyMulDiv}(f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$ $\triangleright$ Algorithm 329: **end if**30: $\mathcal{Q}.\text{append}(Q), \mathcal{R}.\text{append}(R)$ $\triangleright$ Final $\mathcal{Q}$ and $\mathcal{R}$ accumulation, skip f 31: **return** $\mathcal{Q}, \mathcal{R}$

Algorithm 5 EvalPoly: Evaluate polynomial product minus remainder via Horner’s method

Input: Polynomials $P_1, \dots, P_n, R \in \mathbb{F}_q[x]$ of degree $\leq d$, evaluation point $z \in \mathbb{F}_q$

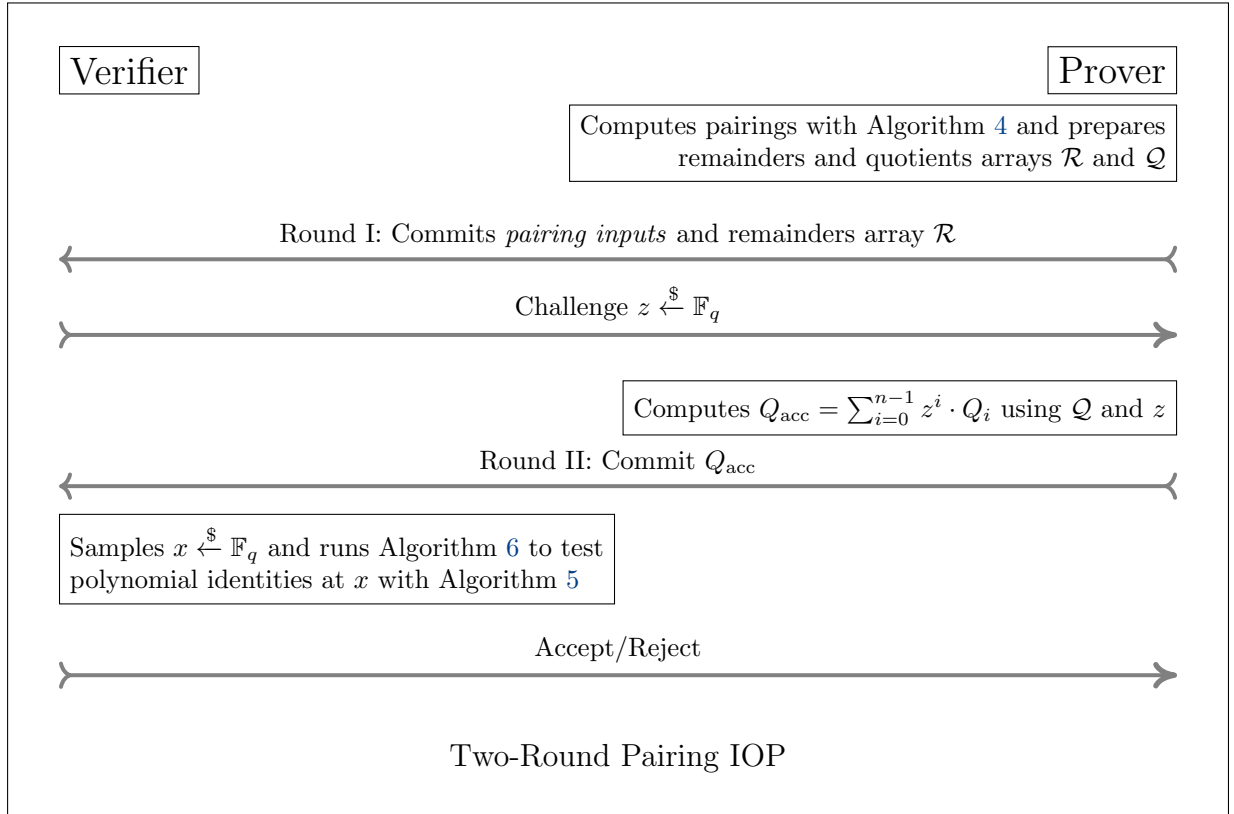
Output: $e \in \mathbb{F}_q$ where $e = \prod_{i=1}^n P_i(z) - R(z)$

```

1: function Eval ( $P = (p_0, p_1, \dots, p_d), z$ ): return (evaluation  $\in \mathbb{F}_q$ )
2:   Uses Horner’s method, body not included for brevity
3:  $e \leftarrow \text{Eval}(P_1, z)$  ▷ Evaluate first polynomial
4: for  $i = 2$  to  $n$  do
5:    $e \leftarrow e \cdot \text{Eval}(P_i, z)$  ▷ Multiply in each subsequent evaluation
6: end for
7:  $e \leftarrow e - \text{Eval}(R, z)$  ▷ Subtract remainder evaluation
8: return  $e$ 

```

4.4 Interactive Proof



4.5 Fiat-Shamir Transformation

Applying the Fiat-Shamir transformation [FS87] with random oracle H gives a non-interactive proof with knowledge soundness $\delta_s + 3(m^2 + 1)2^{-\lambda}$ [BSCS16], where $m = 2$ (hash queries) and $\lambda = 254$ (BN254 hash output length).

Gnark’s multi-commitment mechanism. In gnark, each `api.Commit` call registers committed variables in the constraint system; the two sequential calls in our protocol produce the two challenges z and x via gnark’s multi-round commitment scheme [BSB, BSB23].

Algorithm 6 Pairing Verifier: Evaluating remainders

Input: $P \in \mathbb{G}_1, Q \in \mathbb{G}_2, c \in \mathbb{F}_{q^{12}}, w \in \{0, 1, 2\}, \mathcal{R}$ array of remainders, $Q_{\text{acc}} \in \mathbb{F}_q[x]$

Internal: 27th root of unity, $W, W^2 \in \mathbb{F}_{q^{12}}$, first round challenge, $z \xleftarrow{\$} \mathbb{F}_q$

Output: $\text{pairing}(P, Q) \stackrel{?}{=} 1$

Write $s = 6t + 2 = \sum_{i=0}^{L-1} s_i 2^i$ where $s_i \in \{-1, 0, 1\}$ and $s_L = 1$

- 1: $c^{-1} \leftarrow 1/c$ ▷ Inverse Residue witness
 - 2: $f \leftarrow c^{-1}$ ▷ Initialize with inverse residue witness
 - 3: $T \leftarrow Q$ ▷ T_i for each pair for multi-pairing
 - 4: $x \xleftarrow{\$} \mathbb{F}_q$ ▷ Sample evaluation point from field
 - 5: $e_{\text{acc}} \leftarrow 0, a_{\text{rlc}} \leftarrow 1$ ▷ Evaluation accumulator and random linear combination multiplier
- Every operation involving T, P or Q is repeated for T_i, P_i and Q_i for multi-pairing.

Miller loop bits

- 6: **for** $i = L - 2$ **to** 0 **do**
- 7: $L_{\text{dbl}} \leftarrow l_{T,T}(P), T \leftarrow [2]T$
- 8: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
- 9: **if** $s_i = 0$ **then**
- 10: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, L_{\text{dbl}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
- 11: **else if** $s_i = 1$ **then**
- 12: $L_{\text{add}} \leftarrow l_{T,Q}(P), T \leftarrow T + Q$
- 13: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c^{-1}, L_{\text{dbl}}, L_{\text{add}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
- 14: **else if** $s_i = -1$ **then**
- 15: $L_{\text{add}} \leftarrow l_{T,-Q}(P), T \leftarrow T - Q$
- 16: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, f, c, L_{\text{dbl}}, L_{\text{add}})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
- 17: **end if**
- 18: $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$ ▷ Update random linear combination factor
- 19: **end for**

Miller loop Frobenius correction

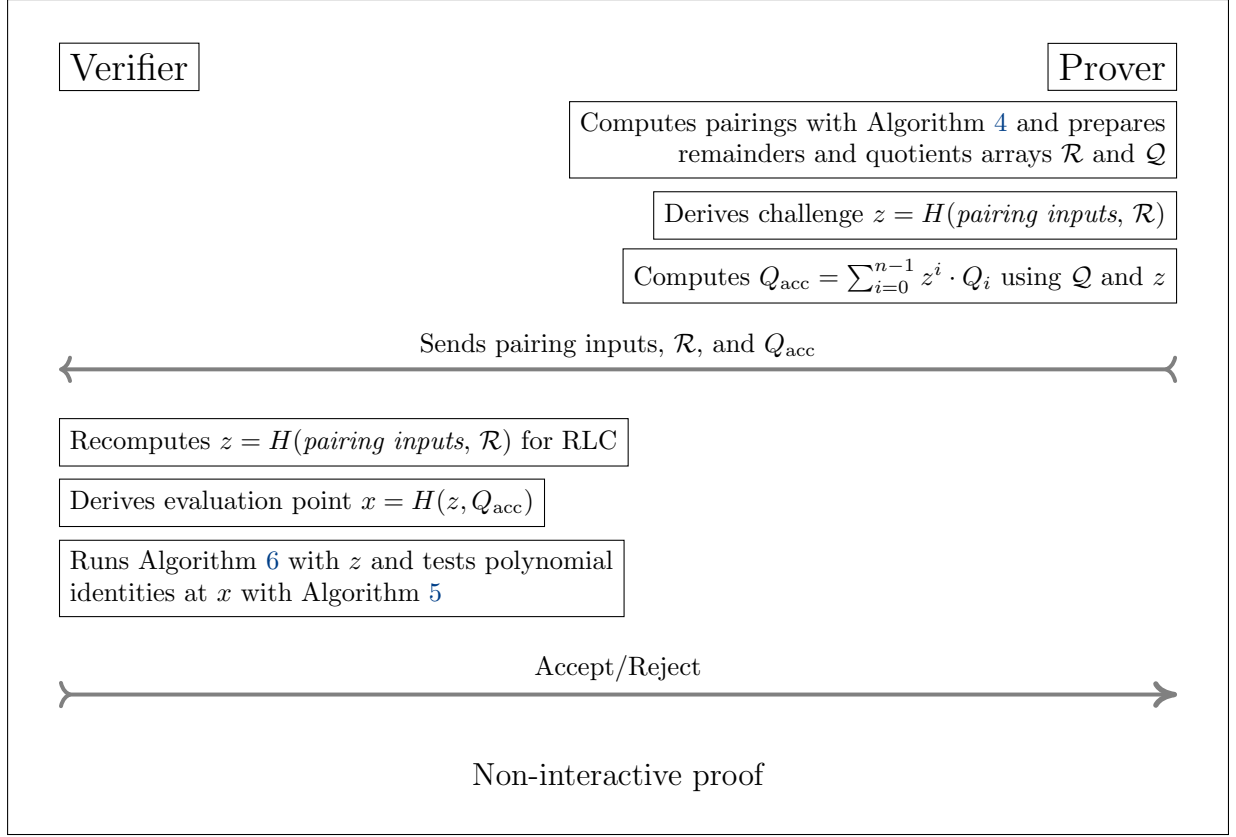
- 20: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
- 21: $Q_1 \leftarrow \pi_p(Q), Q_2 \leftarrow \pi_p^2(Q)$
- 22: $L_{\pi} \leftarrow l_{T,Q_1}(P)$
- 23: $T \leftarrow T + Q_1$
- 24: $L_{-\pi^2} \leftarrow l_{T,-Q_2}(P)$
- 25: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, L_{\pi}, L_{-\pi^2})$ ▷ Algo 5, add L_{*i} for each P_i, Q_i
- 26: $f \leftarrow R, a_{\text{rlc}} \leftarrow a_{\text{rlc}} \cdot z$ ▷ Update random linear combination factor

Final exponentiation: Cubic scaling + Frobenius mappings

- 27: $R \leftarrow \mathcal{R}.\text{pop_front}()$ ▷ Fetch R from prover
- 28: **if** $w = 0$ **then**
- 29: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
- 30: **else if** $w = 1$ **then**
- 31: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
- 32: **else if** $w = 2$ **then**
- 33: $e_{\text{acc}} \leftarrow e_{\text{acc}} + a_{\text{rlc}} * \text{EvalPoly}(R, x, f, W^2, c^{-q}, c^{q^2}, c^{-q^3})$ ▷ Algo 5
- 34: **end if**

- 35: **return** $Q_{\text{acc}}(x) \stackrel{?}{=} e_{\text{acc}} \wedge R \stackrel{?}{=} 1$ ▷ Polynomial identity and expected pairing output check
-

4.6 Non-Interactive Proof



Concrete commitment costs: In gnark, each call to `api.Commit` registers committed variables in the constraint system; the precise prover overhead depends on the backend (Groth16 or Plonk) [BSB, BSB23].

4.7 Security Analysis

Throughout this section, $z, x \in \mathbb{F}_q$ denote field element challenges (scalars), while polynomials are elements of $\mathbb{F}_q[x]$ with x as the formal indeterminate. In particular, z^j is a scalar coefficient, not a polynomial term.

4.7.1 Algebraic Intuition

Before the formal soundness proof it is instructive to consider why an incorrect remainder is difficult to hide. Suppose a malicious prover corrupts exactly one remainder: $R'_j = R_j + \Delta_j$ with $\Delta_j \in \mathbb{F}_q[x]$, $\Delta_j \neq 0$, $\deg(\Delta_j) \leq 11$. Since z is obtained *after* the commitment to $\mathcal{R}$, the prover cannot adjust any other R_k to compensate for Δ_j . To successfully deceive the verifier, the prover must find a valid forged quotient $Q'_{\text{acc}} \in \mathbb{F}_q[x]$ satisfying

$$\sum_{i=0}^{n-1} z^i (A_i - R'_i) = Q'_{\text{acc}} \cdot P_{12}$$

All $R'_i = R_i$ except for $R'_j = R_j + \Delta_j$. Substituting $R'_j = R_j + \Delta_j$ gives

$$\begin{aligned} Q'_{\text{acc}} \cdot P_{12} &= Q_{\text{acc}} \cdot P_{12} - z^j \Delta_j(x) \\ Q'_{\text{acc}} &= Q_{\text{acc}} - \frac{z^j \Delta_j(x)}{P_{12}(x)}. \end{aligned}$$

Here $z^j \in \mathbb{F}_q$ is a nonzero scalar (with overwhelming probability over the choice of z) and $\Delta_j(x) \in \mathbb{F}_q[x]$ is a nonzero polynomial of degree at most 11. Since P_{12} is irreducible of degree 12 and $\deg(z^j \Delta_j) \leq 11 < 12 = \deg(P_{12})$, the polynomial P_{12} does not divide $z^j \Delta_j$ in $\mathbb{F}_q[x]$; hence $z^j \Delta_j / P_{12}$ is not a polynomial. Consequently no valid $Q'_{\text{acc}} \in \mathbb{F}_q[x]$ can satisfy the equation for this single-corruption case.

The same degree argument extends immediately to an adversary who corrupts *multiple* remainders: for any scalar z , the weighted sum $\sum_i z^i \Delta_i(x)$ is still a polynomial in x of degree at most 11 (since each z^i is a scalar and each $\deg(\Delta_i) \leq 11$), so it cannot be divisible by P_{12} of degree 12 unless it is the zero polynomial, which requires every $\Delta_i = 0$. The formal soundness proof below re-derives this uniformly via Schwartz-Zippel.

4.7.2 Soundness

Theorem 1 (Soundness). *Let n be the number of polynomial equations batched via a random linear combination into the bivariate polynomial identity:*

$$P_{\text{final}}(z, x) = \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] - Q_{\text{acc}}(x) \cdot P_{12}(x) = 0$$

Let the maximum degree in x of the polynomial products be $d_x < 2^8$. For a prover committing to incorrect remainders, the probability of successful forgery δ_s over randomly chosen $(z, x) \in \mathbb{F}_q^2$ is bounded by:

$$\delta_s \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

Proof. If the prover commits to an incorrect remainder $R_i(x)$, the evaluation yields a non-zero error polynomial $P_{\text{err}}(z, x)$. The structure of $P_{\text{final}}(z, x)$ establishes the following degree bounds for $P_{\text{err}}(z, x)$:

- **Degree in z :** at most $n - 1$ (due to the z^i weighting).
- **Degree in x :** at most $d_x < 2^8$ (due to the polynomial products).

By the Schwartz-Zippel lemma, the probability that the non-zero polynomial P_{err} evaluates to zero is bounded by the total degree divided by the field size:

$$\Pr[P_{\text{err}}(z, x) = 0] \leq \frac{\deg_z(P_{\text{err}}) + \deg_x(P_{\text{err}})}{|\mathbb{F}_q|} \leq \frac{n - 1 + d_x}{|\mathbb{F}_q|}$$

□

Concrete parameters for BN254: With $n = 67$ equations and $d_x \leq 256$:

$$\delta_s \leq \frac{66 + 256}{2^{254}} < \frac{2^9}{2^{254}} = 2^{-245}.$$

Non-interactive proof via Fiat-Shamir : From Section 4.5, under the random oracle model the knowledge soundness error becomes [BSCS16]:

$$\delta'_s = \delta_s + 3(m^2 + 1) \cdot 2^{-\lambda}$$

where $m = 2$ (hash queries) and $\lambda = 254$ (hash output length), giving

$$\delta'_s \leq 2^{-245} + 15 \cdot 2^{-254} < 2^{-244}.$$

Remark 1 (Correctness of polynomial evaluation). *Theorem 1 assumes that the verifier evaluates all polynomials at the challenge point x without arithmetic error. The circuit implementation uses the deferred-reduction inner-product optimisation of Section 5.5 (Section 5.5.1), which accumulates limb-wise products over $\mathbb{Z}$ before a single final reduction. Correctness of this optimisation requires that no intermediate limb value reaches the native field characteristic p . We verify this bound concretely for BN254 in Lemma 1 (Section 5.5.1).*

4.7.3 Completeness

For an honest prover executing Algorithm 4 with valid inputs:

1. **Polynomial construction:** At each operation, the prover computes following polynomials:

$$\prod_j P_{i,j} = Q_i \cdot P_{12} + R_i$$

2. **RLC formation:** Given challenge z , the prover forms:

$$Q_{\text{acc}} = \sum_{i=0}^{n-1} z^i \cdot Q_i$$

3. **Verifier's check:** The verifier evaluates at point x :

$$\begin{aligned} \text{LHS} &= \sum_{i=0}^{n-1} z^i \cdot \left[\prod_j P_{i,j}(x) - R_i(x) \right] \\ &= \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \cdot P_{12}(x) \quad (\text{by construction}) \\ &= P_{12}(x) \cdot \sum_{i=0}^{n-1} z^i \cdot Q_i(x) \\ &= P_{12}(x) \cdot Q_{\text{acc}}(x) = \text{RHS} \end{aligned}$$

Since the equality holds for any evaluation point x , the protocol satisfies perfect completeness.

5 Polynomial Ring Toolkit

The protocol of Section 4 is packaged as a reusable toolkit in the `emulated` package of gnark [Conb], parameterised by any emulated field $\mathbb{F}_q$ (via `FieldParams` type T) and any modulus polynomial $P \in \mathbb{F}_q[x]$. The toolkit operates in a recursive proof setting where the prover’s computations are performed outside the circuit via `hints` (Section 5.2). And verifier’s checks are implemented as constraints inside the circuit.

Refer to our fork at <https://anonymous.4open.science/r/tiny-gnark-DDCD> for code.

5.1 Ring Representation

A ring element is essentially a coefficient slice `[]*Element[T]` in ascending degree order. A `nil` entry is treated as zero by `InnerProductNoReduce`—the term is skipped entirely—so sparse polynomials carry no wasted constraint variables.

The optional `modEvalFn` of type `EvalFnType[T] = ([*Element[T]]) → *Element[T]` allows handling small-integer coefficients efficiently by multiplying by the small positive integer first and negating afterwards, rather than multiplying by the full-width field representative $q - c$.

Example: BN254 $\mathbb{F}_{q^{12}}$

$P_{12}(x) = x^{12} - 18x^6 + 82$ has nine zero coefficients (stored as `nil`) and two small non-zero ones. The `modEvalFn` computes $P_{12}(\alpha) = \text{MulConst}(\alpha^0, 82) + \text{Neg}(\text{MulConst}(\alpha^6, 18)) + \alpha^{12}$, keeping intermediate values small instead of multiplying by $q - 18$.

5.2 Hint Functions

Hints allow defining computations outside of a circuit. A hint defines an annotated function in the circuit at compile time. Inputs and outputs are injected at the solving time. We define these two hints to implement the prover’s steps in the two-round protocol of Section 4.4.

`polyRingMulHint` implements `POLYMULDIV` (Algorithm 3): it multiplies the input polynomials, divides by P to obtain (Q, R) , and serialises the result as `nbQTerms | Q limbs | nbRemTerms | R limbs`. Inputs are packed as `nbBits | nbLimbs | nbPoly | q | inputs | P`, where each polynomial is prefixed by its term count.

`quotientsRLCHint` computes the accumulated quotient $Q_{\text{acc}}[j] = \sum_i z^i \cdot Q_i[j] \bmod q$ coefficient-wise, corresponding to the prover’s step in Algorithm 4.

5.3 API

The toolkit exposes three entry points. `NewPolyRingCheck` registers a ring group (modulus P plus an optional small-coefficient evaluator) and returns a handle shared by every multiplication under that modulus; multiple groups with different moduli may coexist. `MulPolyRings` computes $R = (\prod_j A_j) \bmod P$ outside the circuit via

`polyRingMulHint`, range-checks R 's limbs inside, and enqueues the check $(A_j; R; Q)$ in the group; for longer product chains a `PolyRingAccumulator` wrapper automatically triggers intermediate reductions when the accumulated degree exceeds a configurable bound. At circuit finalisation, `performDeferredRingChecks` implements Equation (5) via two sequential `api.Commit` calls [BSB] (producing challenges z and x) and one `AssertIsEqual` per group, so no in-circuit work beyond remainder range-checks is performed until this point. Full API signatures and a step-by-step code walkthrough are available in the repository.²

5.4 Usage

Core API

Register the ring group once at initialisation:

```
fp, _ := emulated.NewField[emulated.BN254Fp](api)
modPoly := fp.MakePoly([]any{82, 0, 0, 0, 0, 0, -18, 0, 0, 0, 0, 0, 1})
ring := fp.NewPolyRingCheck(modPoly, nil)
// Each Miller-loop step then calls:
rem, _ = f.MulPolyRings(ring, f, f, L1, L2, L3)
```

and `performDeferredRingChecks` is invoked once by the gnark backend at finalisation, running the full two-round protocol of Section 4.

Accumulate a product chain

```
acc := f.NewPolyRingAccumulator(group, targetDeg)
acc.Mul(A)
acc.Mul(B)
acc.Sqr() // enqueues (A·B)2 without multiplying yet
rem := acc.Eval()
```

5.5 Implementation Details and Performance

5.5.1 Schoolbook vs Horner's Method for Polynomial Evaluation

Horner's method for evaluations across emulated elements causes a linear increase in the limbs count per coefficient. This necessitates reductions after each multiplication. Instead, we cache powers of the evaluation point x in $\mathcal{X} = (x^0, x^1, \dots, x^d)$ and evaluate each polynomial as an inner product of $\mathcal{X}$ and the coefficient array $\mathbf{F}$:

$$\mathcal{X} \cdot \mathbf{F} = \mathcal{X}_0 \cdot F_0 + \mathcal{X}_1 \cdot F_1 + \dots + \mathcal{X}_d \cdot F_d$$

In the emulated field context, EVAL in Algorithm 5 is replaced by this inner product with $\mathcal{X}$. This allows us to use the inner product optimization described below.

²<https://anonymous.4open.science/r/tiny-gnark-DDCD>

5.5.2 Native field operations for inner products

Modular reductions can be deferred or entirely elided for operations expressible as inner products, such as polynomial evaluations and random linear combinations. This optimization leverages the capacity headroom provided by representing small emulated limbs (e.g., 64-bit limbs) within a significantly larger native field element.

Deferred Reduction: Intermediate scalars are permitted to grow over the integers $\mathbb{Z}$ during limb-wise multiplications and linear accumulations.

Preservation of Identity: Algebraic identities are strictly preserved provided the maximum accumulated limb magnitude remains bounded below the native characteristic p .

Handling Asymmetry: In circuits with asymmetric multiplicative depths, operands may accumulate as disparate polynomial multiples of the emulated characteristic p_{emul} . Consequently, a terminal single-element reduction suffices for equivalence testing, bypassing the constraint overhead of intermediate normalization.

Lemma 1 (Overflow safety for deferred reduction). *Let $\mathbb{F}_q$ be emulated with limbs of width b bits inside a native field $\mathbb{F}_p$. Consider evaluating a degree- d polynomial via an inner product over the cached power vector $\mathcal{X}$. Since limbs are accumulated independently, each result-limb is a sum of $d+1$ products of two b -bit values, so its magnitude is bounded by $\beta = 2^{2b + \lceil \log_2(d+1) \rceil}$.*

The deferred single-element reduction is safe whenever $\beta < \log_2 p$.

Corollary 1 (BN254 pairing instantiation). *For BN254 pairing verification: $b = 64$, $d \leq 256$, and $p \approx 2^{254}$. Then*

$$\beta = 2 \cdot 64 + \lceil \log_2 257 \rceil = 128 + 9 = 137 < 254,$$

so the deferred reduction is safe.

6 Conclusion

We have shown that choosing the right granularity for polynomial identity testing—entire Miller loop steps rather than individual field multiplications—yields substantial and measurable improvements in arithmetized pairing verification. The key technique, collapsing a composite multi-operand product into a single Euclidean division check with batched quotient accumulation, is simple to implement and broadly applicable to other multi-step cryptographic computations in arithmetized environments.

6.1 Applications and Impact

Our approach delivers maximum impact in resource-constrained execution environments where custom auxiliary data can be provided for cheaper verification. **Recursive proof systems** benefit the most greatly reducing outer circuit complexity from fewer commitments and operations. **ZKVMs** (RISC Zero, SP1, Jolt) benefit naturally as they express computation as constraint systems compatible with our polynomial framework. **Calldata-constrained blockchains** (Ethereum L2s, BitVM) also benefit, as fewer commitments means smaller proof transcripts and associated calldata costs.

Table 2: Performance improvements in Garaga: Cairo implementation on Starknet

MULMOD		ADDMOD		COMMITTS		VM STEPS	
Before	After	Before	After	Before	After	Before	After
BN254: 3-Pairing with Miller loop and final exponentiation							
19,142	12,656	21,686	15,142	5,689	2,807	338,678	212,902
33.9%		30.2%		50.7%		37.1%	
BLS12-381: 3-Pairing with Miller loop and final exponentiation							
16,484	11,287	20,669	15,420	5,421	3,167	325,757	219,155
31.5%		25.4%		41.6%		32.7%	
MULMOD/ADDMOD: Field operations in $\mathbb{F}_p$ via built-in function calls							
COMMITTS: Use transaction calldata and Poseidon hash operations for Fiat-Shamir							
VM STEPS: CairoVM Steps [Sta22] (CPU Cycles)							

This is already implemented in-production in Garaga [FE] for CairoVM and is currently the standard for all pairing based ZK proofs on Starknet. The polynomial ring toolkit 5 in gnark, submitted as a part of this work, will directly impact `evmprecompiles/ecpair` precompile via `AssertMillerLoopAndFinalExpIsOne`.

6.2 Implementation and Performance

The Garaga library [FE] provided an earlier proof of concept for this technique in Cairo/Starknet (2024). The resulting reductions in commitment overhead and modulo operations (Table 2) were sufficient to fit Groth16 proof verification within Starknet’s transaction limits for the first time. However, Cairo [Sta22] is a specialized environment with its own cost metrics, making direct comparison with mainstream ZK frameworks difficult. This work situates the same technique within gnark—the standard framework for field emulation in arithmetized circuits—providing a systematic benchmark against an industry baseline.

Table 3 presents benchmark results from our gnark implementation of the polynomial ring toolkit, measured on a Groth16 verification simulation on BN254. The end-to-end results confirm both directions of saving: **39% SCS constraint reduction** ($1,890,771 \rightarrow 1,158,194$) and **60% commitment reduction** ($739,198 \rightarrow 295,946$), with strong secondary improvements across compile-time memory ($\sim 42\text{--}43\%$) and solver memory ($\sim 26\%$ SCS, $\sim 52\%$ R1CS).

Benchmark circuit. Both implementations are evaluated against the same gnark circuit³,

Step 1: A 2-pair fixed- $\mathbb{G}_2$ pairing check $e(P_1, Q_1) \cdot e(P_2, Q_2) = 1$, where both $\mathbb{G}_2$ points use precomputed line evaluations (`PairingCheck` with fixed Q).

Step 2: A single in-circuit Miller loop $ML(P_3, Q_3)$ combined with a Miller loop result f computed *outside* the circuit, followed by a final-exponentiation-equals-one assertion (`AssertMillerLoopAndFinalExpIsOne`).

This structure simulates constraints for a real Groth16 verification: the two fixed- $\mathbb{G}_2$ pairings, one full pairing, and one $\mathbb{F}_{q^{12}}$ multiplication. These recreate the four pairing operations with fixed trusted setup points.

³Circuit definition: https://archive.org/download/ppprt_groth16_simulation_test

Table 3: Performance comparison: gnark implementation, Groth16 simulation on BN254 (2 fixed-point pairings + 1 full pairing + 1 $\mathbb{F}_{q^{12}}$ mul)

Metric	gnark Baseline	This Work	Reduction
SCS (Plonkish) Constraint System			
Nb Constraints	1,890,771	1,158,194	38.7%
Nb Commitments	739,198	295,946	60.0%
Compile Time	775.64 ms	406.71 ms	47.6%
Compile Memory	1.77 GB	1.03 GB	41.8%
Solve Time	479.61 ms	356.42 ms	25.7%
Solve Memory	510.36 MB	376.57 MB	26.2%
R1CS Constraint System			
Nb Constraints	588,013	353,881	39.8%
Nb Commitments	739,198	295,946	60.0%
Compile Time	780.40 ms	423.09 ms	45.8%
Compile Memory	2.09 GB	1.19 GB	43.1%
Solve Time	320.84 ms	214.73 ms	33.1%
Solve Memory	330.54 MB	158.51 MB	52.0%

Commitments: Fiat-Shamir `api.Commit` calls. Hardware: Apple M3 Pro (`goarch:arm64`).

Circuit benchmarked: https://archive.org/download/ppprt_groth16_simulation_test

Our implementation: <https://anonymous.4open.science/r/tiny-gnark-DDCD/>

The **gnark baseline** numbers were obtained by running the benchmark function against the upstream gnark `emulated` package (github.com/consensys/gnark). The **This Work** numbers were obtained from the same benchmark file compiled against our fork, which replaces the per-multiplication deferred checks in the `emulated` package with the polynomial ring toolkit described in this paper. No changes to the circuit definition or test harness were required; only the underlying `emulated` arithmetic backend differs between the two runs.

6.3 Further Work

Several natural extensions of this work remain open. First, our framework could be extended to additional curves: **BLS12–381** for applications requiring higher security than BN254, and **BW6–761** for recursive proof chain verification on Ethereum.

A further direction is exploring the technique for native field pairings, such as BLS12–377 inside BW6–761 [Hou23] scalar fields, where the trade-off between batching savings and commitment costs warrants dedicated analysis.

Rather than targeting a specific pairing, we designed this as a general polynomial ring toolkit, which opens additional avenues for exploration. One promising direction is larger batches that fold multiple Miller loop steps into a single polynomial verification. This strategy works well in Garaga implementation. But is not pursued in our implementation because in Gnark implementation polynomial evaluations are cheap, while the extra evaluation challenge powers required to handle degree doubling from cross-step squarings require in place reductions. Finally, the underlying technique generalizes beyond pairings to any setting involving repeated operations over the same polynomial rings in arithmetized environments.

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